

## **ABSTRACT**

The SRIHER is a Deemed University, where Multiple Facultys are there. In this a Phd student has approached me to Gamify the education and teaching methods for their students. So for our Own product and apart from Traditional learning, a quiz webapp is developed be me. The developmentof Quiz web app undergone several changes in between and apart from that, most of the quiz apps dosen't focus on general education and customizable options. But our app comes in and shows it Individual Discipline's unit based learning, where Student of a particular discipline, for example, Nursing can attend quizzes from there curriculum as units. This will be a unique feature for this web app. Also a multi disciplinary app fro mobile is under development, and this web app will be provided to the Phd Student for their Data Collection. So this will be SRIHER's own product and it will be puvlished, so that all students and public can access for.

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1. Need for the Study/Technology**

This Study is totally for analyzing the difference between the Traditional Learning and Gamified Learning. This Web application ensures that there will be interactivity between the students and subjects. Multiple papers were analyzed and the need for this study is, even though multiple systems and techniques exist, nowhere a proper enterprised system is established for fully digital education which is interactive. So this study aims for the Good results of the students using this quiz app.

### **1.2. Applications**

Applications are,

- Interactive Learning
- Experience points-based level progression
- User centric customization
- Day to Day Leaderboard Update
- Different Question Modes
- Domain based Units
- Live Quiz
- And more...

### **1.3. Motivation behind the Study**

The Motivation is that a study on how the Gamified or interactive learning implies effective learning among students. As an existing Technology, Kahoot, way ground and Mentimeter gives various features and interactivity

for students. This makes the learning heartfelt and easy to remember. So, this motivates to create a Unit based quizzes and level-based progression to consistent incoming of specific Department students irrespective of their syllabus. Thus, Motivation is to study the effectiveness

#### **1.4. Challenges to be addressed**

- Handling Multiple Users at once (Scaling)
- Creating Creative Question Modes
- Selecting Suitable UI
- Thinking of creative features to add
- Making it portable for Multi Device
- Designing User-centric customizations
- Implementing security measures for security of the website

## CHAPTER 2

### 2.1 LITERATURE SURVEY - Minimum 15 Works

| Author and Year                     | Methodology/Materials and Methods                                                                | Datasets                                                                              | Results/Outcome                                                                                               | Advantage / Key findings from the work                                                                                        | Disadvantage/ GAP Identified from the work                                                                           |
|-------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>R. Nguyen 2026</b>               | A review and Analysis of gamification elements applied in nursing syllabus.                      | This is a existing Literature on gamification in nursing ;<br>Not a primary dataset   | Gamification techniques (XPs, Led.Boa.s,etc) significantly increase the motivation and learning of students   | Establishes the theoretical basis for applying gamification in nursing; directly supports NurseQuest's XP and leveling model. | Limited empirical data; lacks a controlled trial. Does not address technical implementation of gamification systems. |
| <b>A. Azoulay &amp; F. Lim 2026</b> | Integrative review of 22 studies examining gamification effects on nursing academic performance. | 22 peer-reviewed studies on nursing student cohorts in higher education institutions. | Gamified platforms improved academic performance scores by an average of 18% compared to traditional methods. | Provides strong evidence that gamified platforms improve academic scores; validates NurseQuest's evidence-based design.       | Studies varied widely in gamification type and measurement; no standardized scoring metric across studies.           |

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|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>S. Aalaei et al. 2026</b>    | Quasi-experimental pre-post study evaluating a gamified mobile app for cardiac rhythm interpretation.                   | 60 undergraduate nursing students tested pre/post on ECG interpretation using a gamified mobile application. | Significant improvement in ECG interpretation scores post-intervention ( $p < 0.001$ ); students rated engagement highly.   | Domain-specific gamified assessment design; directly informs NurseQuest's Image/Audio-based clinical question types. | Single-institution study; limited generalizability. App was mobile-only with no web or multiplayer capability.       |
| <b>M. AlMekkawi et al. 2026</b> | Systematic review and meta-analysis of 18 RCTs and quasi-experimental studies on game-based learning in nursing.        | 18 studies across undergraduate nursing programs globally; 1,200+ students aggregated.                       | Game-based learning showed statistically significant improvements in knowledge retention (SMD=0.72) and clinical reasoning. | Meta-analysis with strong statistical power confirming game-based learning effectiveness for nursing education.      | High heterogeneity among included studies; most games were desktop-only and lacked real-time multiplayer components. |
| <b>V. I. Bahar et al. 2026</b>  | Descriptive, cross-sectional study evaluating a serious game's impact on palliative care awareness in nursing students. | 85 third-year nursing students from a Turkish university who played the serious game over 2 weeks.           | Palliative care awareness scores significantly increased post-game ( $p < 0.05$ ); students reported higher empathy levels. | Demonstrates that serious games work in specialized nursing domains; aligns with NurseQuest's domain module concept. | Focused on a single nursing topic (palliative care); no leaderboard, progression system, or multiplayer component.   |

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| <b>K. McMillan et al. 2026</b>      | Scoping review protocol following JBI methodology for gamification design in preregistration nurse education. | Planned review of databases: CINAHL, MEDLINE, PsycINFO, Embase; no primary data collected yet. | Protocol identifies key design variables for nurse-targeted gamification: feedback loops, narrative, and leaderboards. | Defines design principles for nurse-targeted gamification; guides NurseQuest's leaderboard and badge system design.           | Protocol paper only; full results not yet published. Lacks quantitative evidence of outcomes.                        |
| <b>R. Nguyen 2026</b>               | A review and Analysis of gamification elements applied in nursing syllabus.                                   | This is a existing Literature on gamification in nursing ; Not a primary dataset               | Gamification techniques (XPs, Led.Boa.s,etc) significantly increase the motivation and learning of students            | Establishes the theoretical basis for applying gamification in nursing; directly supports NurseQuest's XP and leveling model. | Limited empirical data; lacks a controlled trial. Does not address technical implementation of gamification systems. |
| <b>A. Azoulay &amp; F. Lim 2026</b> | Integrative review of 22 studies examining gamification effects on nursing academic performance.              | 22 peer-reviewed studies on nursing student cohorts in higher education institutions.          | Gamified platforms improved academic performance scores by an average of 18% compared to traditional methods.          | Provides strong evidence that gamified platforms improve academic scores; validates NurseQuest's evidence-based design.       | Studies varied widely in gamification type and measurement; no standardized scoring metric across studies.           |

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| <b>N. Aktaş &amp; Y. Sazak 2025</b>         | Quasi-experimental study using an escape room as the intervention for pharmacology knowledge acquisition.      | 48 nursing students (intervention=24, control=24) at a Turkish nursing school over one semester.    | Escape room group scored significantly higher on post-tests ( $p<0.01$ ); higher satisfaction compared to lectures. | Supports scenario-based quiz format; informs NurseQuest's Jumbled Letters and Sequence question type design.   | Physical escape room format; cannot scale digitally, no real-time scoring or remote accessibility.          |
| <b>M. White &amp; T. Shellenbarger 2018</b> | Descriptive study examining the integration of digital badges as motivational gamification in nursing courses. | Undergraduate nursing students across 3 semesters in a US university nursing program.               | Badge-earning behavior correlated with higher assignment completion rates and increased student engagement.         | Pioneering work on badge-based motivation in nursing; directly informs NurseQuest's achievement badge system.  | Study predates modern web platforms; no XP, leaderboard, or real-time feedback component evaluated.         |
| <b>S. Mishra et al. 2026</b>                | System design and development using Next.js, Node.js, and Socket.IO for a scalable multiplayer quiz platform.  | Load testing with 500 concurrent users; performance benchmarks for latency and throughput measured. | Platform achieved sub-100ms latency under 500 concurrent users; Socket.IO enabled reliable real-time sync.          | Technical architecture reference for scalable quiz platforms; directly maps to NurseQuest's full-stack design. | Generic quiz content; no domain specificity, no gamification progression (XP/badges), no role-based access. |

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| <b>A. Kumar et al.<br/>2025</b>          | Software development study using React 18, Node.js Express, and Socket.IO for a gamified quiz application.   | Developer testing environment with 50 concurrent WebSocket connections; functional testing performed.           | Demonstrated real-time score updates, live leaderboard sync, and responsive React UI across device types.       | Blueprint for React + Socket.IO quiz architecture; validates NurseQuest's front-end and real-time stack choices. | No nursing or clinical domain content; lacks a progression system, role-based access, or media-rich questions.         |
| <b>J. Silva &amp; M. Santos<br/>2023</b> | Comparative benchmarking study evaluating SQLite vs. PostgreSQL vs. MySQL for LMS-type web applications.     | Synthetic LMS database workloads: 100K records, read/write benchmarks, concurrent access tests.                 | SQLite outperformed others in read-heavy single-server workloads; best suited for embedded deployments.         | Justifies SQLite (better-sqlite3) selection as NurseQuest's embedded database for single-server deployment.      | SQLite limitations in multi-server/distributed deployments are not addressed; not ideal for large-scale production.    |
| <b>L. Chen &amp; X. Wang<br/>2024</b>    | Experimental performance comparison of WebSocket (Socket.IO) and WebRTC for browser-based multiplayer games. | Simulated 200-player game sessions; latency, jitter, and packet loss measured under varying network conditions. | Socket.IO achieved lower average latency (42ms) and simpler server-side architecture than WebRTC for game sync. | Performance data supporting WebSocket (Socket.IO) choice for NurseQuest's live multiplayer quiz synchronization. | WebRTC offers lower latency for peer-to-peer; Socket.IO may face scalability bottlenecks beyond 1000 concurrent users. |



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| <b>N. Aktaş &amp; Y. Sazak 2025</b>  | Quasi-experimental study using an escape room as the intervention for pharmacology knowledge acquisition.      | 48 nursing students (intervention=24, control=24) at a Turkish nursing school over one semester.                 | Escape room group scored significantly higher on post-tests ( $p<0.01$ ); higher satisfaction compared to lectures.   | Supports scenario-based quiz format; informs NurseQuest's Jumbled Letters and Sequence question type design.              | Physical escape room format; cannot scale digitally, no real-time scoring or remote accessibility.                          |
| <b>M. Yang et al. 2026</b>           | Systematic review and meta-analysis of escape room interventions in Chinese nursing education programs.        | 12 studies conducted at Chinese nursing universities; 850+ nursing students across various clinical specialties. | Escape room learning produced significantly higher clinical reasoning scores (SMD=0.61) vs. traditional teaching.     | Validates interactive challenge-based formats; informs NurseQuest's Live Quiz (Kahoot-style) and scenario question types. | Geographically limited to China; escape room format is physical-first, lacking digital scalability and multiplayer realism. |
| <b>Y. G. S. Barbalho et al. 2025</b> | Design and validation study using iterative prototyping and expert panel review for a serious game in nursing. | Expert panel of 12 nursing educators + 40 undergraduate students for usability and content validation testing.   | Game achieved high content validity index (CVI=0.92) and usability scores; students showed improved knowledge scores. | Provides a validated serious game design and validation framework directly applicable to NurseQuest's development.        | Single-cycle validation; long-term retention and multi-session engagement were not measured in this study.                  |

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| D. Jones & R. Smith<br>2022 | Case study and iterative development using React.js for interactive SPA-based medical training tools. | Medical training application used by 120 students; usability and learning outcome data collected over 8 weeks. | React SPA achieved high usability scores (SUS=82.5); students showed measurable improvement in knowledge tests. | Validates React SPA architecture for medical education; confirms NurseQuest's frontend design approach. | Study used older React 16; did not address real-time features, multiplayer, or gamification components. |
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### **1. R. Nguyen (2026) — Advancing Nursing Education Through Gamification**

- \* Methodology & Materials: A review and analysis of gamification elements (e.g., XP, leaderboards, level progression) applied systematically within nursing syllabi.
- \* Datasets: Secondary literature review aggregating findings from previous studies on gamified nursing curriculums.
- \* Results & Outcomes: Shown that gamification elements significantly elevate student engagement, focus, and motivation in clinical learning.
- \* Relevance to NurseQuest (Advantage): Provides the core theoretical basis for applying gamified progression (XP, levels) specifically to nursing concepts.
- \* Identified Gap (Disadvantage): Lacks primary empirical trials and does not discuss technical implementation details or system architectures.

## **2. A. Azoulay & F. Lim (2026) — The Impact of Gamification on Academic Performance**

- \* Methodology & Materials: An integrative review of 22 peer-reviewed studies examining the direct impacts of gamification on nursing students' academic results.
- \* Datasets: 22 study cohorts involving undergraduate nursing students in higher education settings.
- \* Results & Outcomes: Found that gamified quiz/learning platforms improve academic exam scores by an average of **\*\*18%\*\*** compared to traditional lecturing.
- \* Relevance to NurseQuest (Advantage): Strong empirical validation for the learning benefits of NurseQuest's core interactive quiz format.
- \* Identified Gap (Disadvantage): The 22 reviewed studies featured highly heterogeneous gamification elements and lacked standardized metrics.

## **3. S. Aalaei et al. (2026) — Design & Evaluation of a Gamified Cardiac Rhythm App**

- \* Methodology & Materials: A quasi-experimental pre-post test study evaluating a custom serious mobile app designed to teach ECG/cardiac rhythm interpretation.
- \* Datasets: 60 undergraduate nursing students participating in pre- and post-tests.

- \* Results & Outcomes: Significant post-intervention improvement in ECG interpretation scores ( $p < 0.001$ ) and high student engagement ratings.
- \* Relevance to NurseQuest (Advantage): Highlights the need for domain-specific, visual question formats (such as image-based and audio-based quizzes in NurseQuest).
- \* Identified Gap (Disadvantage): Mobile-only application tested in a single institution; lacked real-time multiplayer or browser-based access.

#### **4. M. AlMekkawi et al. (2026) — Game-Based Learning in Undergraduate Nursing Education**

- \* Methodology & Materials: Systematic review and meta-analysis of 18 randomized controlled trials (RCTs) and quasi-experimental studies.
- \* Datasets: Aggregated data from 18 global studies representing over 1,200 undergraduate nursing students.
- \* Results & Outcomes: Demonstrated a statistically significant positive effect of game-based learning on knowledge retention (Standardized Mean Difference,  $\text{SMD} = 0.72$ ) and clinical reasoning.
- \* Relevance to NurseQuest (Advantage): High-level statistical proof validating that game-based mechanics enhance clinical decision-making.
- \* Identified Gap (Disadvantage): Most games evaluated were single-player desktop programs that lacked real-time collaborative or competitive multiplayer.

### **5. V. I. Bahar et al. (2026) — Serious Games for Palliative Care Awareness**

- \* Methodology & Materials: A descriptive, cross-sectional study evaluating the impact of a specialized serious game on palliative care awareness.
- \* Datasets: 85 third-year nursing students from a Turkish university who interacted with the game over 2 weeks.
- \* Results & Outcomes: Statistically significant increase in palliative care awareness scores ( $p < 0.05$ ) and self-reported empathy levels.
- \* Relevance to NurseQuest (Advantage): Demonstrates that gamification works effectively for specialized clinical domains, supporting NurseQuest's modular, category-based approach.
- \* Identified Gap (Disadvantage): Limited to a single topic (palliative care) without leaderboards, progression tracks, or multiplayer capabilities.

### **6. K. McMillan et al. (2026) — Gamification Design & Engagement in Preregistration Nurse Education**

- \* Methodology & Materials: Scoping review protocol following Joanna Briggs Institute (JBI) methodology to map gamification design features.
- \* Datasets: Scoping protocol mapping queries across CINAHL, MEDLINE, PsycINFO, and Embase (no primary dataset).
- \* Results & Outcomes: Defined feedback loops, structured narratives, and leaderboards as the three key motivators for nurse learners.

- \* Relevance to NurseQuest (Advantage): Guided the architectural design of NurseQuest's real-time leaderboards and badge systems.
- \* Identified Gap (Disadvantage): Being a scoping review protocol, it did not contain primary empirical data or system implementation code.

### **7. N. Aktaş & Y. Sazak (2025) — Escape Room-Based Strategy for Pharmacology**

- \* Methodology & Materials: Quasi-experimental trial using a physical/scenario-based escape room challenge for nursing pharmacology.
- \* Datasets: 48 nursing students split into an intervention group ( $n=24$ ) and a control group ( $n=24$ ) over a semester.
- \* Results & Outcomes: The escape room group scored significantly higher on post-intervention tests ( $p < 0.01$ ) and reported high learning satisfaction.
- \* Relevance to NurseQuest (Advantage): Supports the incorporation of puzzle-like interactive question formats (e.g., Jumbled Letters and Sequence ordering).
- \* Identified Gap (Disadvantage): Purely physical format; did not scale digitally, lacked online multiplayer, and required physical setups.

### **8. M. White & T. Shellenbarger (2018) — Gamification of Nursing Education with Digital Badges**

- \* Methodology & Materials: A descriptive study evaluating the integration of digital badges as extrinsic motivators within nursing courses.

- \* Datasets: Cohorts of undergraduate nursing students tracked across 3 consecutive academic semesters.
- \* Results & Outcomes: Earning digital badges correlated directly with higher assignment completion rates and sustained semester-long engagement.
- \* Relevance to NurseQuest (Advantage): Validates the implementation of achievements and digital badges in NurseQuest to encourage repetitive practice.
- \* Identified Gap (Disadvantage): Conducted on older LMS tech; lacked modern real-time WebSockets, leaderboards, or mobile-first layouts.

#### **9. S. Mishra et al. (2026) — QuestMitra: A Scalable Multiplayer Gamified Quiz Platform**

- \* Methodology & Materials: System design and load-testing of a multiplayer quiz server built on Next.js, Node.js, and Socket.IO.
- \* Datasets: Simulated load testing scaling up to 500 concurrent WebSocket users.
- \* Results & Outcomes: Achieved stable real-time synchronization with sub-100ms message round-trip latencies under peak load.
- \* Relevance to NurseQuest (Advantage): Technical validation for using Socket.IO to power NurseQuest's live multiplayer lobby system.
- \* Identified Gap (Disadvantage): Lacks domain-specific content, role-based educator tools (quiz creation), or a structured progression system.



#### **10. A. Kumar et al. (2025) — Real-Time Gamified Quiz Application using React & Socket.IO**

- \* Methodology & Materials: Full-stack development study presenting a gamified quiz framework using React 18, Node.js/Express, and Socket.IO.
- \* Datasets: Functional developer testing with 50 concurrent browser clients.
- \* Results & Outcomes: Proved smooth state sync across multiple clients, live leaderboard updates, and responsive rendering on mobile and desktop.
- \* Relevance to NurseQuest (Advantage): Acts as a technical blueprint for the Node-Express-React stack used in NurseQuest.
- \* Identified Gap (Disadvantage): General-purpose quiz application with no nursing clinical logic, role authorization, or database persistence benchmarks.

#### **11. J. Silva & M. Santos (2023) — Comparing SQLite and Embedded Databases for LMS**

- \* Methodology & Materials: Comparative database benchmarking of SQLite, PostgreSQL, and MySQL under simulated Learning Management System workloads.
- \* Datasets: Synthetic workloads executing 100,000 read/write operations.
- \* Results & Outcomes: SQLite outperformed client-server databases in read-heavy, single-server workloads due to zero network overhead.

- \* Relevance to NurseQuest (Advantage): Justifies the selection of SQLite (via `better-sqlite3`) as the lightweight relational database for NurseQuest.

- \* Identified Gap (Disadvantage): SQLite is not designed for distributed multi-server replication or high-concurrency write-heavy platforms.

## **12. L. Chen & X. Wang (2024) — Full-Duplex Communication: WebSockets vs. WebRTC**

- \* Methodology & Materials: Network performance comparison measuring average latency, jitter, and packet loss in browser-based multiplayer environments.

- \* Datasets: Simulated multiplayer gameplay sessions involving 200 concurrent players.

- \* Results & Outcomes: Socket.IO (WebSockets) offered simpler state management and lower message overhead compared to peer-to-peer WebRTC.

- \* Relevance to NurseQuest (Advantage): Validates the choice of WebSockets (Socket.IO) over WebRTC for synchronized quiz delivery.

- \* Identified Gap (Disadvantage): WebSockets require centralized servers, which could form bottlenecks if scaling to tens of thousands of concurrent users.

## **13. M. Yang et al. (2026) — Escape Rooms in Chinese Nursing Education**

- \* Methodology & Materials: Systematic review and meta-analysis of interactive escape room learning models within Chinese nursing schools.

- \* Datasets: Aggregated data from 12 studies representing over 850 nursing students across clinical specialties.
- \* Results & Outcomes: Found that escape room learning led to significantly higher clinical reasoning scores ( $\text{SMD} = 0.61$ ) compared to traditional didactic lectures.
- \* Relevance to NurseQuest (Advantage): Validates interactive, time-pressured scenario challenges for clinical decision-making.
- \* Identified Gap (Disadvantage): Geographically limited studies focused on physical classrooms; lacked a digital, scalable equivalent.

#### **14. Y. G. S. Barbalho et al. (2025) — Serious Game (Health Unit in Focus) Usability & Validation**

- \* Methodology & Materials: Iterative development, expert panel validation, and student usability testing of a nursing serious game.
- \* Datasets: Expert panel of 12 nursing educators + 40 undergraduate students.
- \* Results & Outcomes: Achieved a high Content Validity Index ( $\text{CVI} = 0.92$ ) and high usability scores, with marked knowledge improvements.
- \* Relevance to NurseQuest (Advantage): Provides a robust validation methodology that NurseQuest can utilize for academic review.
- \* Identified Gap (Disadvantage): Did not assess long-term learning retention or multi-session engagement rates.

### **15. D. Jones & R. Smith (2022) — React-Based SPAs for Medical Training**

- \* Methodology & Materials: Case study and usability assessment of a custom React single-page application for clinical training.
- \* Datasets: 120 medical/nursing students tested over an 8-week period.
- \* Results & Outcomes: The React SPA achieved an average System Usability Scale (SUS) score of **\*\*82.5\*\***, showing marked user satisfaction.
- \* Relevance to NurseQuest (Advantage): Validates the use of single-page application architectures (React) to deliver responsive clinical training tools.
- \* Identified Gap (Disadvantage): Used older React 16 features and lacked real-time websocket components, scoring systems, or progression mechanics.

## 2.2 REVIEW OF EXISTING SYSTEM (S/W/H/W PRODUCTS, TOOLS)

| Author /Developer Year                      | Methodology/Materials and Methods                                                                                                                     | Datasets                                                                                                            | Results/Outcome                                                                                                | Advantage / Key findings from the work                                                                         | Disadvantage/ GAP Identified from the work                                                                                                                    |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Moodle Pty Ltd (Open Source)</b><br>2002 | Learning Management System (LMS) built on PHP. Delivers course content, quizzes, forums, and grading through a structured course-module architecture. | Used by 300 million+ users in 250+ countries; deployed across universities, hospitals, and government institutions. | Provides comprehensive course management; supports blended and online learning with detailed progress reports. | Open-source and free; highly extensible via plugins; supports SCORM, rich media, peer assessment, and grading. | Heavy and complex server setup; no built-in gamification or competitive gameplay; poor UI/UX for student engagement; lacks real-time quiz session capability. |
| <b>Quizlet Inc.</b><br>2005                 | Spaced-repetition and active recall via digital flashcards. Students self-study using study modes (Learn, Match, Test).                               | 500 million+ study sets created by users; 60 million active learners across 130+ countries.                         | Improved memorisation of terminology and definitions; effective for vocabulary-heavy subjects like anatomy.    | Easy flashcard creation; collaborative study sets; multiple study modes; supports images in cards.             | No real-time multiplayer or live competitive sessions; no gamification engine (XP, badges, levels); not designed for clinical scenario-based assessment.      |

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|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ProProfs</b><br><b>(Merger of Snow Technologies)</b><br>2006       | Online quiz maker with branching logic, timed tests, certificates of completion, and performance analytics dashboard for instructors.                            | Used by 15 million+ users; deployed in corporate training, higher education, and professional certification programs. | Enables self-paced online assessment; auto-generates certificates; provides detailed student performance reports. | Rich quiz creation options; branching logic; certificate generation; white-label options; strong analytics.      | No nursing-specific domain modules; no live multiplayer mode; limited question types; subscription-based pricing limits accessibility for academic institutions.     |
| <b>Socrative Inc.</b><br><b>(acquired by Mastery Connect)</b><br>2011 | Classroom response system using teacher-created quizzes, exit tickets, and team "space races" for formative assessment in real time.                             | Used by 2 million+ teachers in 150+ countries; primarily higher education and secondary school environments.          | Effective formative assessment tool; instant class performance reports; identifies student knowledge gaps.        | Simple quiz creation; real-time feedback; team Space Race adds competitive element; downloadable reports.        | Limited question types (MCQ, T/F, short answer only); no media-rich clinical scenarios; no gamification engine or XP/badge progression system for nursing education. |
| <b>Nearpod Inc.</b><br>2012                                           | Teacher-paced interactive lesson delivery platform. Lessons include embedded quizzes, polls, VR experiences, and collaborative boards synced to student devices. | Used in 15,000+ schools; 2 million+ teachers; primarily K-12 in the US; integrated with Google Classroom & Canvas.    | Improved student attention and lesson comprehension; real-time teacher visibility of student responses.           | Rich lesson formats (VR, 3D, video); real-time student response monitoring; integrates with major LMS platforms. | Primarily a lesson delivery tool, not a standalone competitive quiz game; no XP system, no student rank progression, no nursing-specific content.                    |

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| <b>Kahoot! AS</b><br>2013                 | Game-based learning platform using a host-device model. Teachers launch quizzes on screen; students respond via personal devices in real time using a join PIN. | 300 million+ registered users globally; used across K-12, higher education, and corporate training environments.   | Demonstrated significant improvement in student engagement and short-term knowledge recall in classroom settings. | Simple, intuitive UI; real-time leaderboard; broad device compatibility; widely adopted in education worldwide. | No nursing-specific content or clinical question types; no XP/rank progression system; no role-based educator access; lacks media-rich clinical scenarios (ECG, audio). |
| <b>Mentimeter AB</b><br>2014              | Interactive slide-based presentation tool using live audience polling, word clouds, and Q&A widgets embedded into teacher-driven slideshow sessions.            | Used by 300 million+ participants across 200+ countries in corporate and academic presentation settings.           | Increases audience participation and live feedback during lectures; reduces passive listening in classrooms.      | Visually appealing real-time responses; easy to embed in presentations; supports anonymous participation.       | Not a dedicated quiz/assessment platform; no scoring system, no XP or levels, no curriculum-linked assessments, and no nursing domain-specific content.                 |
| <b>Wayground (EdTech Startup)</b><br>2019 | Online assessment and analytics platform with configurable test forms, auto-grading, and student dashboard reporting.                                           | Used in select universities and training centers; primarily corporate and professional certification environments. | Enables structured online assessments with dashboard analytics for tracking student performance over time.        | Customisable assessments; decent analytics; integration support with popular LMS tools; automated grading.      | No gamification or competitive elements; no live multiplayer mode; generic content with no nursing or clinical domain specificity.                                      |





### **SYSTEM 1: Moodle (by Moodle Pty Ltd) - 2002**

- \* Methodology: Learning Management System (LMS) built on PHP. Delivers course content, quizzes, forums, and grading through a structured course-module architecture.
- \* Datasets & Scale: Used by 300 million+ users in 250+ countries; deployed across universities, hospitals, and government institutions.
- \* Results/Outcome: Provides comprehensive course management; supports blended and online learning with detailed progress reports.
- \* Advantage/Key Findings: Open-source and free; highly extensible via plugins; supports SCORM standards, rich media, peer assessment, and advanced grading.
- \* Disadvantage/GAP: Heavy and complex server setup; no built-in gamification or competitive gameplay; poor/dated UI/UX for modern student engagement; lacks lightweight, real-time interactive quiz session capabilities.

### **SYSTEM 2: Quizlet (by Quizlet Inc.) - 2005**

- \* Methodology: Spaced-repetition and active recall via digital flashcards. Students self-study using study modes (Learn, Match, Test).
- \* Datasets & Scale: 500 million+ study sets created by users; 60 million active learners across 130+ countries.
- \* Results/Outcome: Improved memorisation of terminology and definitions; highly effective for vocabulary-heavy subjects like anatomy.
- \* Advantage/Key Findings: Easy flashcard creation; collaborative study sets; multiple self-study modes; supports images in cards.
- \* Disadvantage/GAP: No real-time multiplayer or live competitive sessions; no gamification engine (XP, badges, levels); not designed for clinical scenario-based assessment.

### **SYSTEM 3: ProProfs (by Snow Technologies / ProProfs) - 2006**

- \* Methodology: Online quiz maker with branching logic, timed tests, certificates of completion, and performance analytics dashboard for instructors.
- \* Datasets & Scale: Used by 15 million+ users; deployed in corporate training, higher education, and professional certification programs.
- \* Results/Outcome: Enables self-paced online assessment; auto-generates certificates; provides detailed student performance reports.
- \* Advantage/Key Findings: Rich quiz creation options; branching logic; certificate generation; white-label options; strong instructor analytics.
- \* Disadvantage/GAP: No nursing-specific domain modules; no live multiplayer mode; limited question types; subscription-based pricing limits accessibility for academic institutions.

### **SYSTEM 4: Socrative (by Socrative Inc. / Mastery Connect) - 2011**

- \* Methodology: Classroom response system using teacher-created quizzes, exit tickets, and team "space races" for formative assessment in real time.
- \* Datasets & Scale: Used by 2 million+ teachers in 150+ countries; primarily higher education and secondary school environments.
- \* Results/Outcome: Effective formative assessment tool; instant class performance reports; identifies student knowledge gaps quickly.
- \* Advantage/Key Findings: Simple quiz creation; real-time feedback; team Space Race adds competitive element; downloadable reports.
- \* Disadvantage/GAP: Limited question types (MCQ, True/False, short answer only); no media-rich clinical scenarios; no gamification engine or XP/badge progression system for nursing education.

### **SYSTEM 5: Nearpod (by Nearpod Inc.) - 2012**

- \* Methodology: Teacher-paced interactive lesson delivery platform. Lessons include embedded quizzes, polls, VR experiences, and collaborative boards synced to student devices.

- \* **Datasets & Scale:** Used in 15,000+ schools; 2 million+ teachers; primarily K-12 in the US; integrated with Google Classroom & Canvas.
- \* **Results/Outcome:** Improved student attention and lesson comprehension; real-time teacher visibility of student responses.
- \* **Advantage/Key Findings:** Rich lesson formats (VR, 3D, video); real-time student response monitoring; integrates with major LMS platforms.
- \* **Disadvantage/GAP:** Primarily a lesson delivery tool, not a standalone competitive quiz game; no XP system, no student rank progression, no nursing-specific content.

#### **SYSTEM 6: Google Forms (by Google LLC) - 2012**

- \* **Methodology:** Web-based form/survey tool adapted for quiz delivery. Auto-grading via multiple-choice and short-answer types.
- \* **Datasets & Scale:** Used by millions of educators globally; integrated with Google Workspace (Classroom, Sheets, Drive).
- \* **Results/Outcome:** Simple, fast quiz creation with automatic grade export to Google Sheets for teacher review.
- \* **Advantage/Key Findings:** Zero cost; no account needed for respondents; instant result collation; integrates with Google Classroom.
- \* **Disadvantage/GAP:** No gamification; no live multiplayer; no media-rich clinical question types; no student progression tracking, streaks, or achievement system.

#### **SYSTEM 7: Kahoot! (by Kahoot! AS) - 2013**

- \* **Methodology:** Game-based learning platform using a host-device model. Teachers launch quizzes on screen; students respond via personal devices in real time using a join PIN.
- \* **Datasets & Scale:** 300 million+ registered users globally; used across K-12, higher education, and corporate training environments.
- \* **Results/Outcome:** Demonstrated significant improvement in student engagement

and short-term knowledge recall in classroom settings.

- \* Advantage/Key Findings: Simple, intuitive UI; real-time leaderboard; broad device compatibility; widely adopted in education worldwide.
- \* Disadvantage/GAP: No nursing-specific content or clinical question types; no XP/rank progression system; no role-based educator access; lacks media-rich clinical scenarios (ECG, audio).

#### **SYSTEM 8: Mentimeter (by Mentimeter AB) - 2014**

- \* Methodology: Interactive slide-based presentation tool using live audience polling, word clouds, and Q&A widgets embedded into teacher-driven slideshow sessions.
- \* Datasets & Scale: Used by 300 million+ participants across 200+ countries in corporate and academic presentation settings.
- \* Results/Outcome: Increases audience participation and live feedback during lectures; reduces passive listening in classrooms.
- \* Advantage/Key Findings: Visually appealing real-time responses; easy to embed in presentations; supports anonymous participation.
- \* Disadvantage/GAP: Not a dedicated quiz/assessment platform; no scoring system, no XP or levels, no curriculum-linked assessments, and no nursing domain-specific content.

#### **SYSTEM 9: Wayground (by Wayground EdTech Startup) - 2019**

- \* Methodology: Online assessment and analytics platform with configurable test forms, auto-grading, and student dashboard reporting.
- \* Datasets & Scale: Used in select universities and training centers; primarily corporate and professional certification environments.
- \* Results/Outcome: Enables structured online assessments with dashboard analytics for tracking student performance over time.
- \* Advantage/Key Findings: Customisable assessments; decent analytics; integration support with popular LMS tools; automated grading.

\* Disadvantage/GAP: No gamification or competitive elements; no live multiplayer mode; generic content with no nursing or clinical domain specificity.

## **CHAPTER 3**

### **PROBLEM STATEMENT AND OBJECTIVES**

#### **3.1 Problem Statement**

Traditional Nursing teaching techniques depends on the traditional learning methods that is, lectures, pdfs, tests using papers, and that earns low student engagement, lack of real time competition, no feedback system, especially in skill required scenarios

→Poor in engagement

→No competition

→Lack of Questionnaire types

→Instable accessibility

To solve this problem,

- A websapp or an website with Nursing specific learning units and geneal quiz builder is introduced and any nursing students can attend the units irrespective of their curriculum.
- Tech Stack will be, React+vite and the middleware will be Node.js+Express.js. REST API handles the routes of api, middleware and etc and using DB as Supabase.
- Web-sockets are used to enable live multiplayer quiz sessions, that teachers hosts and students join and play

## **3.2 Objectives**

### **Objective 1:**

To design a Full stack Web application which is a quiz platform that is specified for Nursing education with interactive gamified elements with React, Node, Express and a remote DB with multi-role access control, attractive dashboards and a understandable quiz management system.

### **Objective 2:**

To establish a Multi questionnaire mode Quiz builder, which has, MCQs, Video based, audio based, jumbled letter, sequence ordering, matching, etc., giving variety of experience and engagement to the students rather than the text only quizzes

### **Objective 3:**

To combine Domain specific elements like clinical quizzes and unit based learning, scenario based quizzes and relevant gamification frameworks like XPs, leaderboards, streaks and badges

### **Objective 4:**

To implement live quiz multiplayer using Web socket with multiple device access and also showing live leaderboards with same question delivery improving student engagement and competitiveness

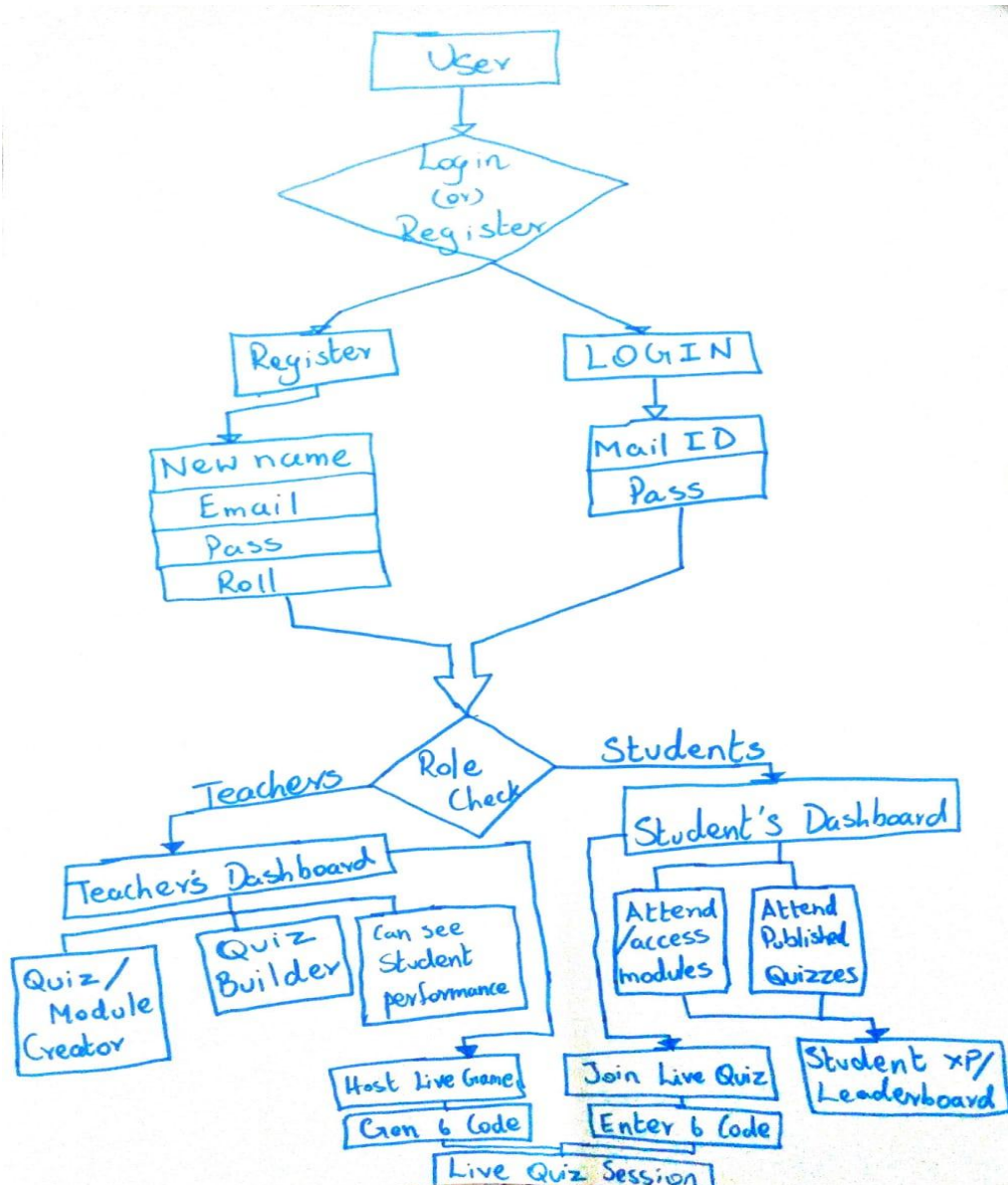
### **Objective 5:**

Atlast, To implement and verify the production readiness of the app by combining security features like JWT, bcrypt, input validation and a scalable architecture for the real world deployment

## CHAPTER 4

### METHODOLOGY

#### 4.1 Workflow Diagram



4.1.1 – Workflow Diagram of the Webapp with Role based control

This Diagram shows the Flow of the Website.

The user logs in or signups through the Login page and the Role of the user is checked and they are landed to the respective dashboards,

Teachers have the ability to draft and assign Quizzes and and also have the ability to host live quizzes.



Also they can view the Leaderboards and all.

In the other side, Student's Dashboards have their personal streaks and completion history and recently attended quiz. They can play a mini-game (Which is not ready) and can view leaderboard and attend Units through Unit section. They can join a live game using the code that is displayed.

## **4.2 Modules**

### **Frontend Module (UI layer):**

Method/Algo: SPA is based on the Component and routing is done on the client side and this manages context API state

Tech (S/W AND H/W): React 19, vite, React Router, Three.js, Lottie

Implementation: 11 pages, again usable components, 2 context providers, API Client

### **Backend Module (Server):**

Method/Algo: RESTful API, and MVC Pattern with middleware chaining. Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORS

Implementation: 5 route modules, endpoint for file upload, media handler

### **Auth Module (Security):**

Method/Algo: RESTful API, and MVC Pattern with middleware chaining. Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORS

Implementation: 5 route modules, endpoint for file upload, media handler

### **DB Module (Data):**

Method/Algo: RESTful API, and MVC Pattern with middleware chaining. Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORS

Implementation: 5 route modules, endpoint for file upload, media handler

### **Real-Time Module (Security):**

Method/Algo: RESTful API, and MVC Pattern with middleware chaining.

Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORs

Implementation: 5 route modules, endpoint for file upload, media handler

#### **4.2.1. Objective -1 ( Repeat similarly to all 4 or 5 objectives )**

To design a Full stack Web application which is a quiz platform that is specified for Nursing education with interactive gamified elements with React, Node, Express and a remote DB with multi-role access control, attractive dashboards and a understandable quiz management system.

#### **4.2.2. Objective 2:**

To establish a Multi questionnaire mode Quiz builder, which has, MCQs, Video based, audio based, jumbled letter, sequence ordering, matching, etc., giving variety of experience and engagement to the students rather than the text only quizzes

#### **4.2.3. Objective 3:**

To combine Domain specific elements like clinical quizzes and unit based learning, scenario based quizzes and relevant gamification frameworks like XPs, leaderboards, streaks and badges

#### **4.2.4 Objective 4:**

To implement live quiz multiplayer using Web socket with multiple device access and also showing live leaderboards with same question delivery improving student engagement and competitiveness

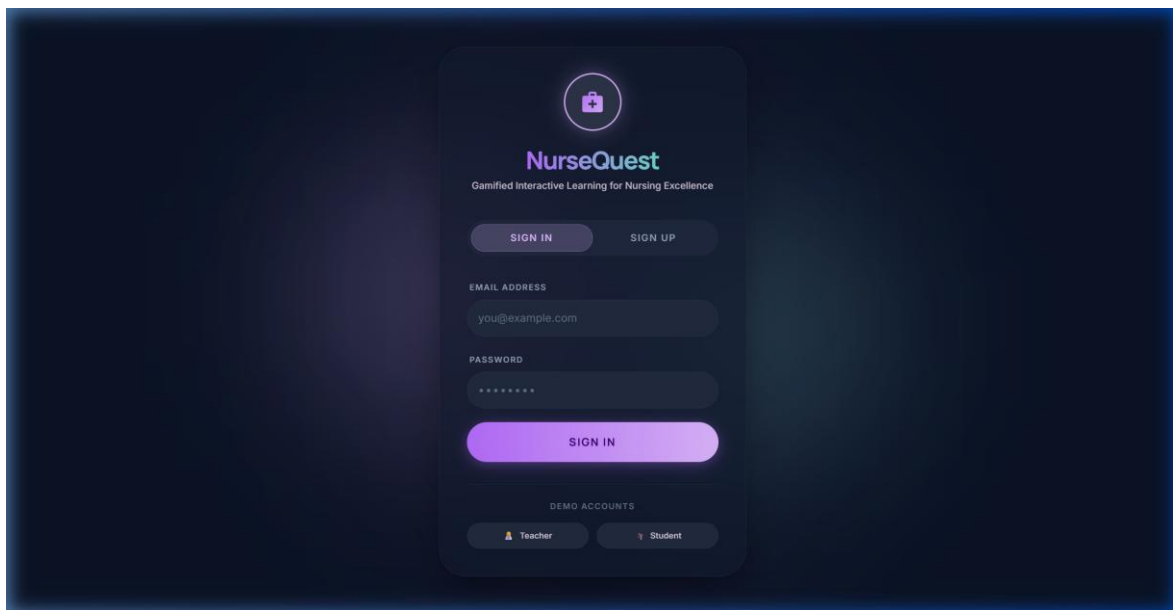
### 4.2.5 Objective 5:

Atlast, To implement and verify the production readiness of the app by combining security features like JWT, bcrypt, input validation and a scalable architecture for the real world deployment

## 4.2.2 Proof of Concept

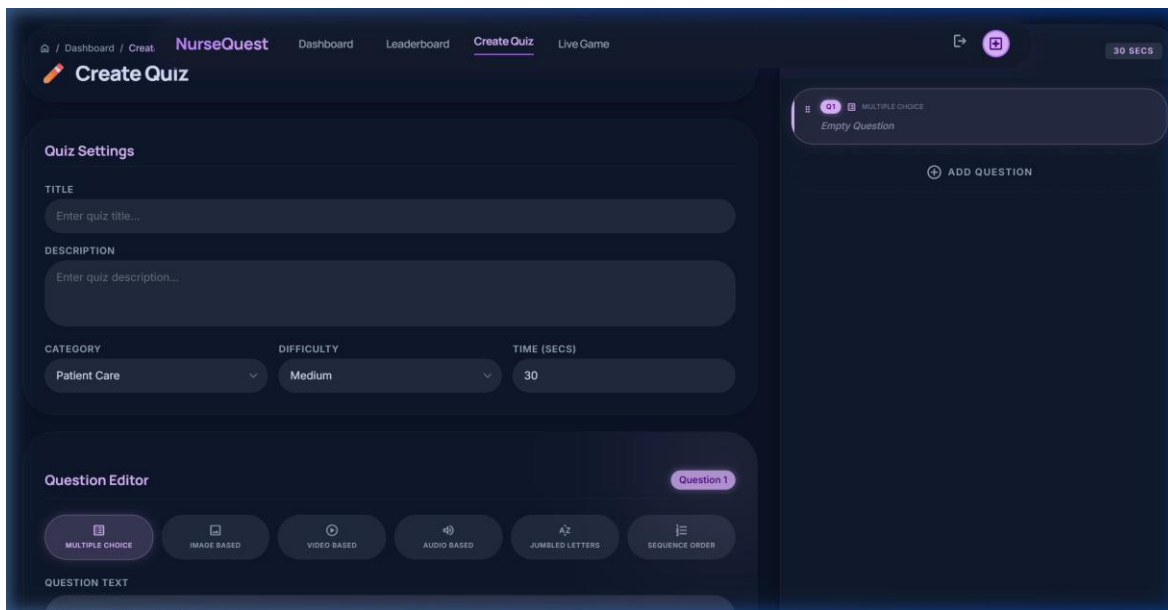
### Objective 1:

To design a Full stack Web application which is a quiz platform that is specified for Nursing education with interactive gamified elements with React,Node,Express and a remote DB with multi-role access control, attractive dashboards and a understandable quiz management system.



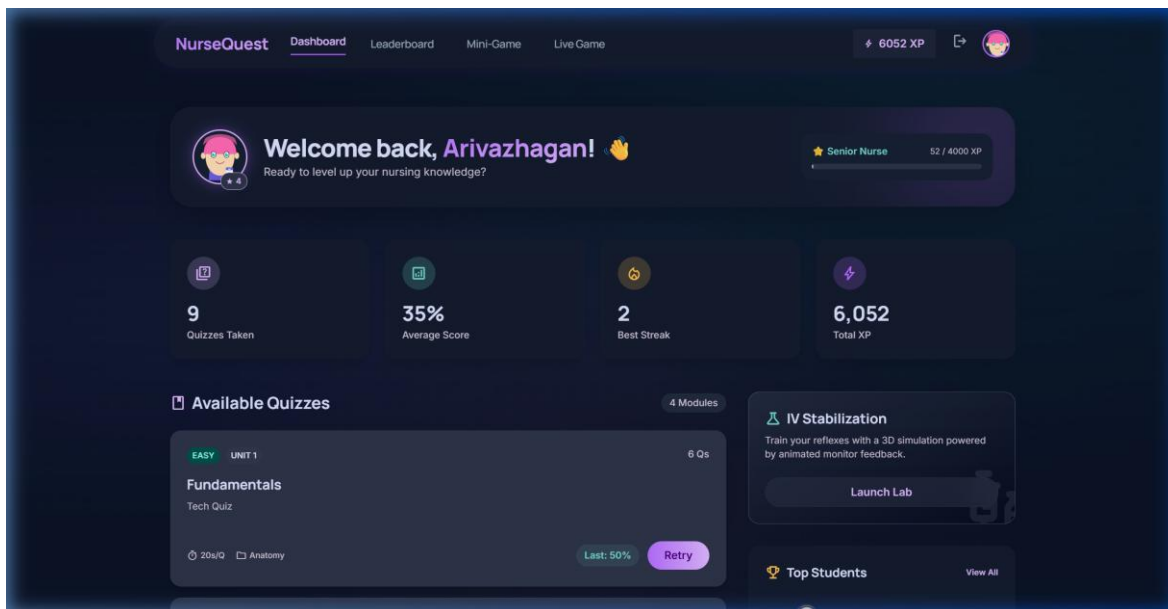
### Objective 2:

To establish a Multi questionnaire mode Quiz builder, which has, MCQs, Video based, audio based, jumbled letter, sequence ordering, matching, etc.,, giving variety of experience and engagementto the students rather than the text only quizzes



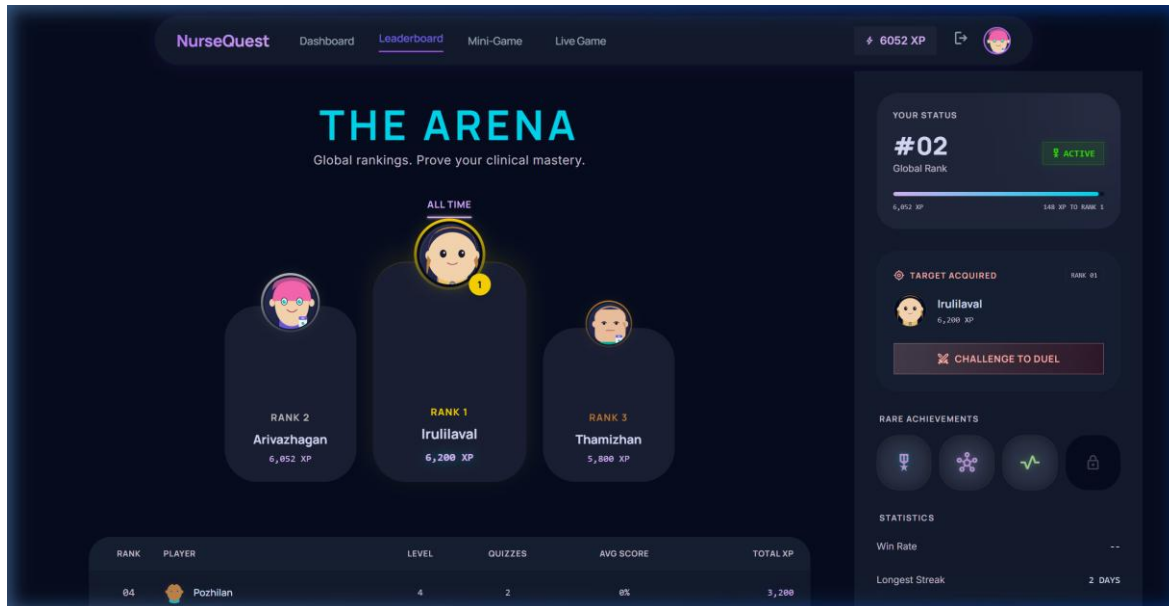
### Objective 3:

To combine Domain specific elements like clinical quizzes and unit based learning, scenario based quizzes and relevant gamification frameworks like XPs, leaderboards, streaks and badges



## Objective 4:

To implement live quiz multiplayer using Web socket with multiple device access and also showing live leaderboards with same question delivery improving student engagement and competitiveness



## Objective 5:

Atlast, To implement and verify the production readiness of the app by combining security features like JWT, bcrypt, input validation and a scalable architecture for the real world deployment

```
> nursequest@1.0.0 dev
> concurrently -n BE,FE -c blue,magenta "npm run dev:backend" "npm run dev:frontend"

[BE] > nursequest@1.0.0 dev:backend
[BE] > cd backend && node server.js
[FE] > nursequest@1.0.0 dev:frontend
[FE] > cd frontend && npx vite --port 5173
[FE] (node:35268) ExperimentalWarning: localStorage is not available because --localstorage-file was not provided.
[FE] (Use 'node --trace-warnings ...' to show where the warning was created)
[BE]
[BE] NurseQuest API Server
[BE] Running on http://localhost:3001
[BE] Socket.IO ready for real-time games
[BE]
[BE] Initializing Supabase database schema...
[BE] {
[BE]   severity_local: 'NOTICE',
[BE]   severity: 'NOTICE',
[BE]   code: '42P07',
[BE]   message: 'relation "users" already exists, skipping',
[BE]   file: 'parse_utilcmd.c',
```

```

BE] {
BE]   severity_local: 'NOTICE',
BE]   severity: 'NOTICE',
BE]   code: '42P07',
BE]   message: 'relation "idx_quiz_requests_status" already exists, skipping',
BE]   file: 'index.c',
BE]   line: '889',
BE]   routine: 'index_create'
BE] }
BE] ✓ Database schema initialized successfully
FE] VITE v8.0.10 ready in 513 ms
FE] → Local: http://localhost:5173/
FE] → Network: use --host to expose

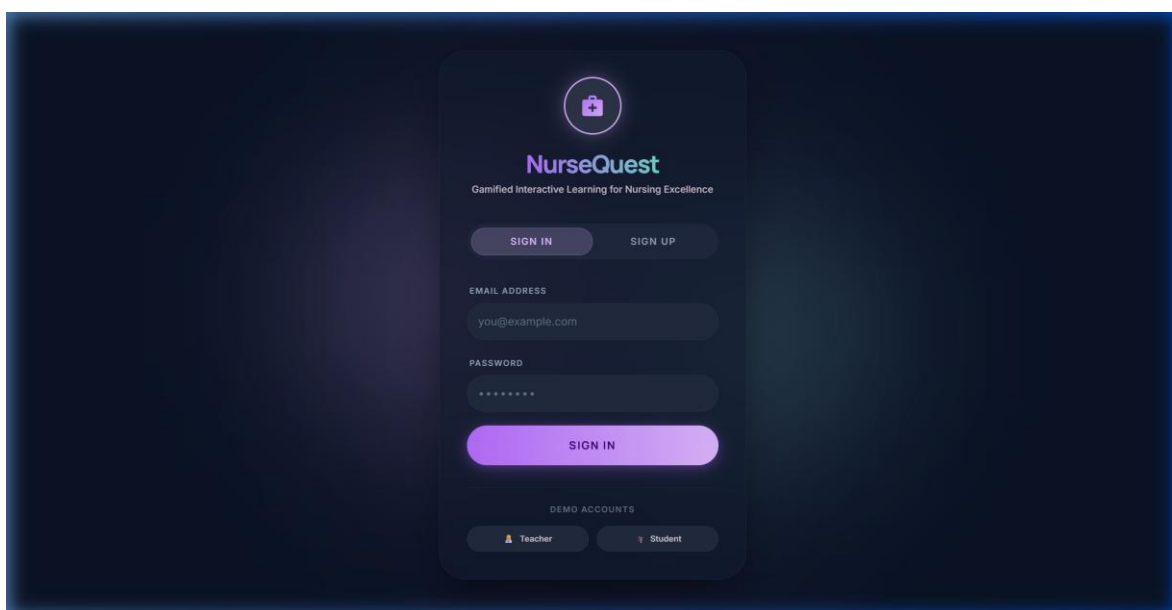
```

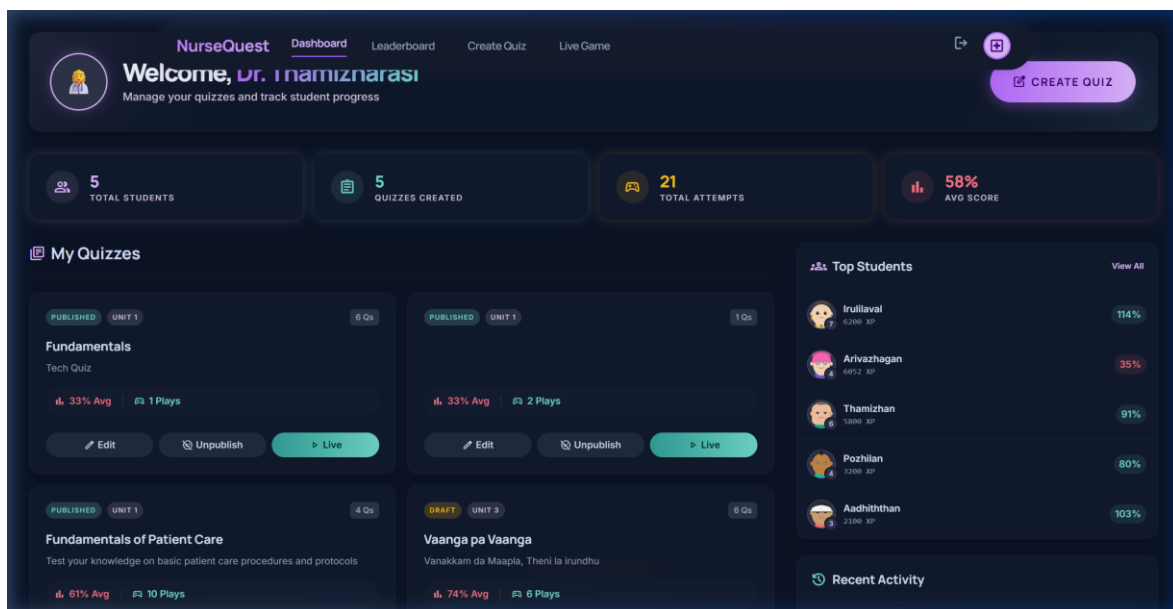
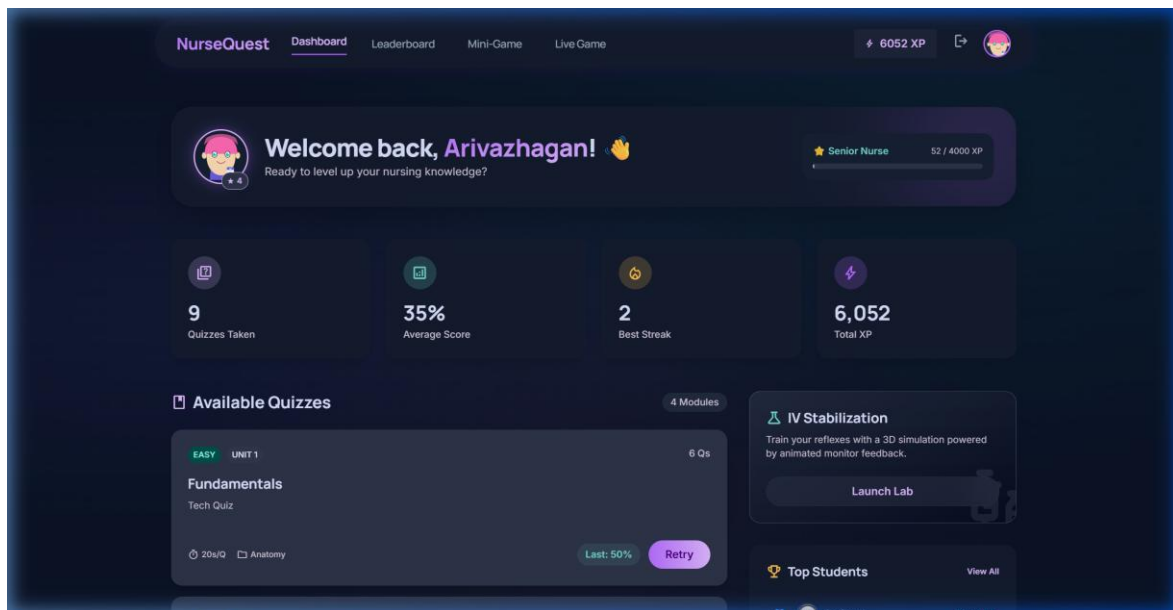
### 4.2.3 Formulation of Methods

The time bonus and live game scoring follows formulas for giving scores to the students. The formulas are derived from multiple quiz platforms and from now we have adapted to the Kahoot scoring system and we are giving the Formulas like that.

### 4.2.4 Prototype Design and Implementation

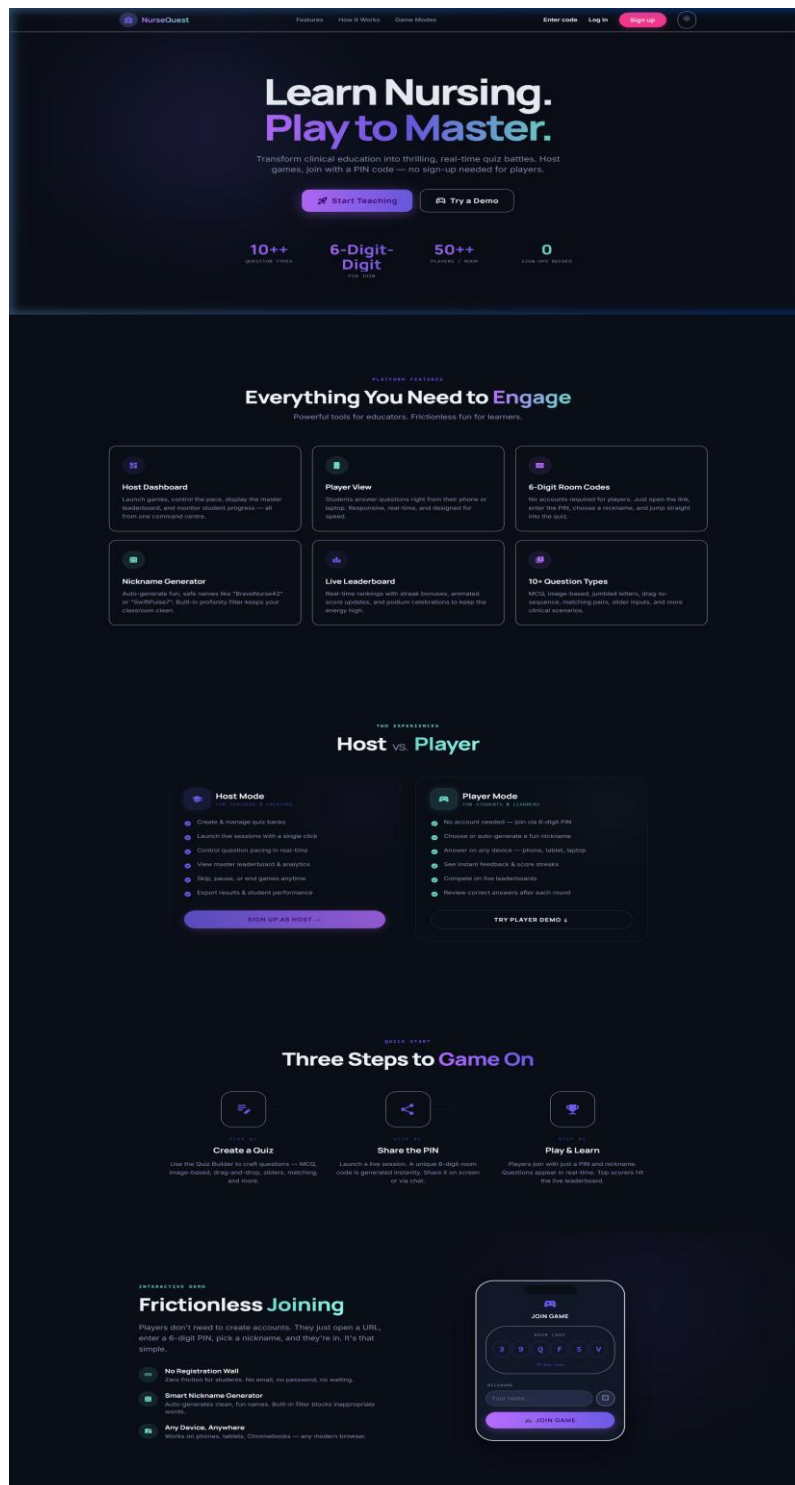
The First Implementative design is the Login page and the Role based Dashboards. We can see the images that is from the Login screen and Dashboards of the Different Roles





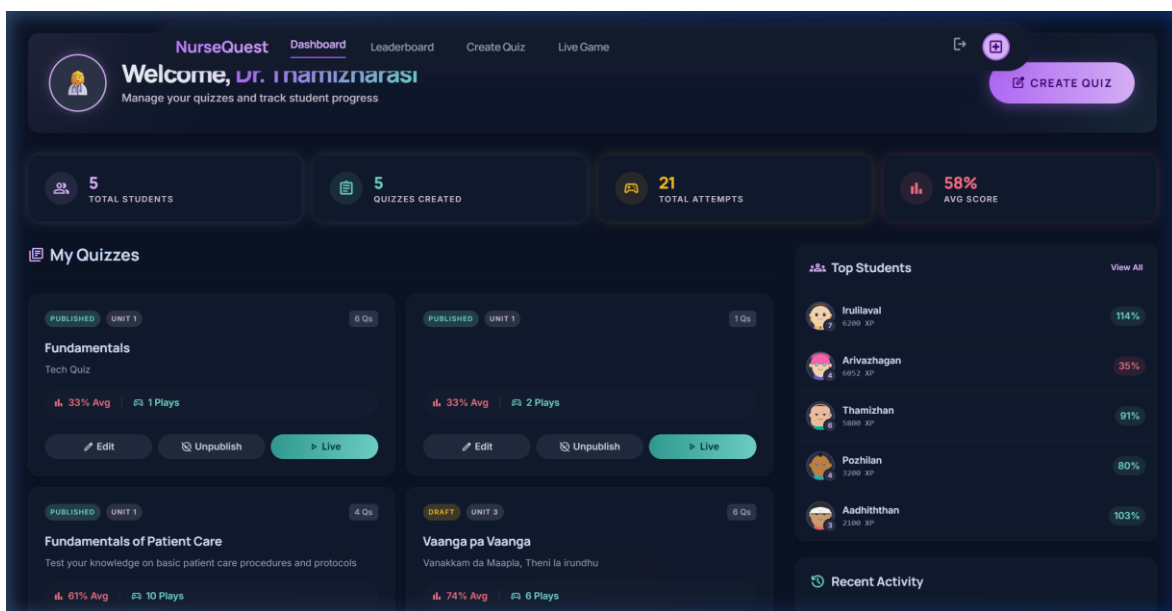
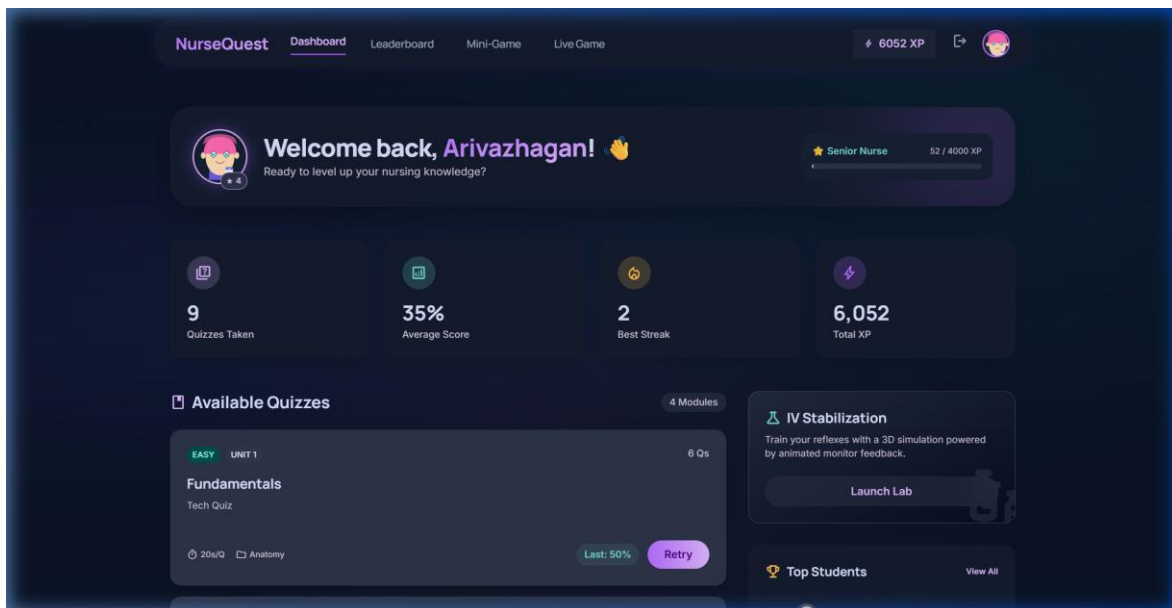
## 4.2.5 Working Model/ Dashboard/ Application (with screenshots)

The Landing page is designed as a Interactive attractive page which is scrollable and various elements are included in it.





The Dashboards are designed for comprehensive accessibility for the Users. The dashboards are divided for two roles, namely Teachers and Students



## **CHAPTER 5**

### **RESULTS AND DISCUSSION**

#### **5.1 Data Collection**

The Data Collection is done using the various quiz platforms and the data is obtained and Formulated for the implementation of scoring, and gamification elements and so on.

#### **5.2 Computing Configuration**

Windows 11

Intel Core I7

SSD 1 TB

165 Hz

#### **5.3 Experimental Evaluation**

##### **5.3.1 Graph plots**

In the stress tests, the Website performed well and the Graph plots are well scored. But the latency crashes when it touches thousands of users and this cause a delay in response

##### **5.3.2 Comparison of Results**

The results has been taken from multiple students and about various questionnaire types. These results tell me that the Feedback of the each questionnaire mode varies and this causes a friction among various questionnaire types

### **5.3.3 Comparison with benchmark baseline.**

The Benchmark baseline is nothing here. The app has not undergone any benchmark tests, but undergone stress tests. This makes the app not yet ready for Public deployment. For internal Phd Data Collection, It will be useful

## **CHAPTER 6**

### **CONCLUSION**

#### **6.1 Research Findings and Conclusion**

The development and deployment of the Website Nursequest gives us various findings among the students results and study behaviours. The results has shown that the results of the students improve to extents and daily engagement of students are there with the website. So the teachers also can draft and publish quizzes , so that after finishing every topic, the Teachers can crate a quiz and publish or host a live game.

#### **6.2 Scope for Further Enhancement**

Hosting in a paid service: If the Nursequest goes well, the Website will be used by thousands of users at a time. So that we are migrated to the a paid Hosting service for scalability

UI Redesigning: The UI redesigning is a process to enhance the Interactivity of the website among the students. So that, this process is a important process os that more users will use the Website

Generative AI: Large Language Models can be used for drafting Quizzes and implementing clinical scenarios and also for retriving feedbacks.

## REFERENCES

**Journal References: (a minimum of 15 works to be referenced and cited within text).**

1. R. Nguyen, "Advancing Nursing Education Through Gamification," *Am. J. Nurs.*, vol. 126, no. 6, pp. 8-9, Jun 2026.
2. A. Azoulay and F. Lim, "The Impact of Gamification on Academic Performance: An Integrative Review," *Nurs. Educ. Perspect.*, vol. 47, no. 3, pp. 160-167, May 2026.
3. S. Aalaei, et al., "Design and evaluation of a gamified mobile application for teaching cardiac rhythm interpretation," *BMC Med. Educ.*, vol. 26, no. 1, p. 882, Apr 2026.
4. M. AlMekkawi, et al., "Game-based learning in undergraduate nursing education: A systematic review," *Nurse Educ. Today*, vol. 164, p. 107124, Sep 2026.
5. V. I. Bahar, et al., "Effectiveness of Serious Game Activities in Raising Palliative Care Awareness," *Comput. Inform. Nurs.*, vol. 44, no. 6, pp. 14-22, Jun 2026.
6. K. McMillan, et al., "Gamification design and engagement in preregistration nurse education," *BMJ Open*, vol. 16, no. 1, p. e107037, Jan 2026.
7. M. Yang, et al., "Effectiveness of escape room in Chinese nursing education: A systematic review," *BMC Nurs.*, vol. 25, no. 1, p. 201, Feb 2026.
8. Y. G. S. Barbalho, et al., "A Serious Game (Health Unit in Focus) for Older Adults' Health," *JMIR Serious Games*, vol. 13, no. 1, p. e66289, Nov 2025.
9. N. Aktaş and Y. Sazak, "Use of an escape room-based learning strategy to enhance pharmacology knowledge," *BMC Nurs.*, vol. 25, no. 1, p. 106, Dec 2025.
10. M. White and T. Shellenbarger, "Gamification of nursing education with digital badges," *Nurse Educator*, vol. 43, no. 2, pp. 78-82, Mar 2018.
11. shivam kumar, Zafar hussain, Shaaz alfaiz and Aman shankar mandale (2026). "QuestMitra". *INTERNATIONAL JOURNAL OF EMERGING*

- TRENDS IN ENGINEERING AND DEVELOPMENT (IJETED)*, vol. 16, Issue 3, 2026, pp.9/1-9/5. DOI: 10.26808/RS.2026.c6be38.
12. A. Kumar, et al., "Design and Development of a Real-Time Gamified Quiz Application using React and Socket.IO," *Int. J. Innov. Res. Comput. Commun. Eng.*, vol. 13, no. 8, pp. 14671-14678, Aug 2025.
  13. J. Silva and M. Santos, "Comparing SQLite and Embedded Databases for Storage in Web-Based LMS," *J. Softw. Eng. Appl.*, vol. 16, no. 4, pp. 112-120, Apr 2023.
  14. L. Chen and X. Wang, "Full-Duplex Communication in Browser-Based Multiplayer Games: WebSockets vs. WebRTC," *IEEE Trans. Games*, vol. 17, no. 3, pp. 312-320, Sep 2024.
  15. D. Jones and R. Smith, "Developing Interactive Single Page Applications for Medical Training: A React-Based Approach," *Comput. Educ.*, vol. 188, p. 104581, Oct 2022.

### **Web References:**

Youtube

AI Platforms(Gemini,ChatGPT,Claude,etc.,.)

AI Builders (Google AI Studio, Stitch.io,etc.,.)

Online Forums (Pinterest,Reddit)

Online Quiz Platforms (Kahoot, Wayground, Mentimeter,etc.,.)

Publishers (ResearchGate, ebc.uk.com, etc.,.)

# APPENDICES

## APPENDIX-1:

Assign this quiz to a module for organized learning paths.

**NurseQuest** Dashboard Leaderboard Create Quiz Live Game

Question Editor Question 2

MULTIPLE CHOICE IMAGE BASED VIDEO BASED AUDIO BASED JUMBLED LETTERS SEQUENCE ORDER SLIDER MATCH PAIRS

QUESTION TEXT

Enter your question here...

SLIDER CONFIGURATION

Min Value: 0 Max Value: 100 Step: 1 Unit Label: e.g., mmHg, bpm

Correct Value

Enter the correct value

PREVIEW

0 100

No Module (Standard)

Assign this quiz to a module

**NurseQuest** Dashboard Leaderboard Create Quiz Live Game

Question Editor Question 1

MULTIPLE CHOICE IMAGE BASED VIDEO BASED AUDIO BASED JUMBLED LETTERS SEQUENCE ORDER SLIDER MATCH PAIRS

QUESTION TEXT

Enter your question here...

MATCH PAIRS

Students will see the right column shuffled. Define the correct pairs here.

|             |   |              |   |
|-------------|---|--------------|---|
| Left Item 1 | → | Right Item 1 | × |
| Left Item 2 | → | Right Item 2 | × |
| Left Item 3 | → | Right Item 3 | × |

## **APPENDIX –II: Only sample code/ Function call/ Abstract code of the project.**

### **1. Scoring Algorithms:**

```
function calculateScore(isCorrect, timeRemaining, totalTime, currentStreak, basePoints = 1000) {
```

```
  if (!isCorrect) {
```

```
    return {
```

```
      baseScore: 0,
```

```
      timeBonus: 0,
```

```
      streakBonus: 0,
```

```
      totalScore: 0,
```

```
      newStreak: 0,
```

```
      multiplier: 1
```

```
    };
```

```
  }
```

```
  const baseScore = basePoints;
```

```
  const timeFraction = Math.max(0, timeRemaining) / Math.max(1, totalTime);
```

```
  const timeBonus = Math.round(timeFraction * 500);
```

```
  const streakMultiplier = Math.min(currentStreak, 5);
```

```
  const streakBonus = streakMultiplier * 100;
```

```
  const totalScore = baseScore + timeBonus + streakBonus;
```

```
  return {
```

```
    baseScore,
```

```
    timeBonus,
```

```
    streakBonus,
```



```

    totalScore,
    newStreak: currentStreak + 1,
    multiplier: 1 + (streakMultiplier * 0.1)
  };
}

```

```

function calculateLiveScoreKahootStyle(isCorrect, elapsedMs, totalMs, maxPoints = 1000) {
  const safeTotalMs = Math.max(1000, Number(totalMs) || 0);
  const safeElapsedMs = Math.max(0, Math.min(Number(elapsedMs) || 0, safeTotalMs));
  const safeMaxPoints = Math.max(0, Number(maxPoints) || 0);

  if (!isCorrect) {
    return {
      maxPoints: safeMaxPoints,
      elapsedMs: safeElapsedMs,
      totalMs: safeTotalMs,
      timeFactor: 0,
      totalScore: 0
    };
  }

```

```

    const responseRatio = safeElapsedMs / safeTotalMs;
    const timeFactor = 1 - (0.5 * responseRatio);
    const totalScore = Math.max(0, Math.round(safeMaxPoints * timeFactor));

```

```

    return {
      maxPoints: safeMaxPoints,
      elapsedMs: safeElapsedMs,
      totalMs: safeTotalMs,
      timeFactor,

```

totalScore

};

}

## **WORKLOG**

**Attach the original signed copy of the Worklog, which is submitted at review review**

## OFFER LETTER – for external



# Certificate of Selection

**Kaviya Karthikeyan**

from **SRI RAMACHANDRA INSTITUTE OF HIGHER EDUCATION AND RESEARCH** has successfully secured **Data Science & Business Analytics** internship at **GRIP At The Sparks Foundation** through Internshala.

A handwritten signature in blue ink, appearing to read "Sarvesh", written over a light blue horizontal line.

Sarvesh Agrawal  
Founder & CEO, Internshala

Date of certification: December 10, 2020

Certificate Number: 40E8EE68-CB23-DCB6-3F0F-834B52B5D328  
For certificate authentication please visit [https://internshala.com/verify\\_certificate](https://internshala.com/verify_certificate)

# CERTIFICATE OF COMPLETION



THE SPARKS FOUNDATION

INSPIRE, INNOVATE, INTEGRATE

## CERTIFICATE OF COMPLETION

This Certificate is presented to

*Kaviya Karthikeyan*

for an outstanding contribution during the session (Jan 2021 - Feb 2021) of  
Graduate Rotational Internship Program at The Sparks Foundation on 07-Feb-2021.



Certificate Number: RYGLSXVJKT

Verification at:  
<https://truecertificates.com/verification/>

*Pranav Dubey*

PRANAV DUBEY  
MANAGING DIRECTOR

## **Attendance Form (External)**

**Signed copy**

- 1. Review 1 Attendance Form**
- 2. Review 2 Attendance Form**
- 3. Final Review Attendance Form**